Thermodynamics of a black hole with cosmological constant in bumblebee gravity

THESIS

Submitted in partial fulfillment

of the requirements for the Master's degree

Sivaraj S Menon

Under the guidance of

Dr. Tharanath.R

Assistant Professor



Department of Physics

St. Xavier's College(Autonomous), Mumbai

2021

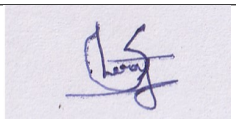
THESIS

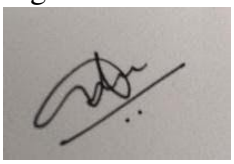
SUBMITTED BY MR.SIVARAJ S MENON

TO **ST. XAVIER'S COLLEGE, MUMBAI**

FOR THE **MASTER'S** DEGREE

IN **PHYSICS**

Name of the Student	Sivaraj S Menon
UID No.	199107
Name of the Guide	Dr. Tharanath.R
Designation and affiliation of Guide	Assistant Professor
Proposed dissertation title	Thermodynamics of a black hole with cosmological constant in bumblebee gravity
Submission date	30 April 2021
Signature of student	

Guide	Supervisor	Head of the Department
Signature 	Signature	Signature
Dr.Tharanath.R, Assistant Professor, Aquinas College, Edacochin, Kerala	Dr. Amruta Sadhu, Postdoctoral Fellow, IIT Gandhinagar	Dr.Jyoti Singh, Associate Professor, Department of Physics, St. Xavier's College, Mumbai.

UNDERTAKING BY THE STUDENT

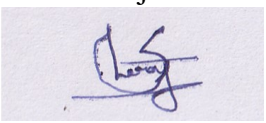
I declare that the work embodied in the thesis entitled **'Thermodynamics of a black hole with cosmological constant in bumblebee gravity'** represents my own contribution to the research carried out under the supervision of **Dr. Tharanath.R** from **Aquinas College.**

The work presented in this thesis has not been submitted for the award of any other degree to this or any other institute. References to other research studies have been duly acknowledged and included in the bibliography.

The thesis has been checked for plagiarism using the software Turnitin, and the soft copy of the report is being submitted along with this thesis.

Date : 30 April 2021

Sivaraj S Menon



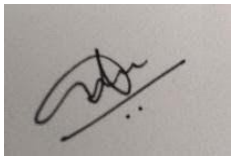
CERTIFICATE FROM THE GUIDE

I certify that the thesis entitled **'Thermodynamics of a black hole with cosmological constant in bumblebee gravity'** submitted by **Mr.Sivaraj S Menon** for the award of Master's degree, embodies original work done by him under my supervision.

This work has not been submitted to this or any other institute for the award of any other degree.

Date : 30 April 2021

Tharanath.R



Assistant Professor

Acknowledgement

I might want to put on record my thankfulness to my specialization where Dr.Jyoti Singh, present Head of the Department , my Professors Dr.Amruta Sadhu, Prof.Katherine Rawlins, and remaining Physics faculties in my college who have all empowered me chasing this work and gave all essential regulatory assistance in finishing the work.

I accept this open door to offer my genuine thanks to my guide Dr. Tharanath R, albeit a couple of words can't do equity to the essential job he has had in my research work. I'm obliged to him for his direction and backing over the span of my work prompting the Project. He showed me with astuteness, consolation, rich information, knowledge, and a profound comprehension of Physics, and all the more critically the best approach to lead scientific research. I'm exceptionally appreciative that he has been consistently accessible to examine the numerous issues and questions I brought him.

I might want to communicate my profound appreciation to every one of my friends. I'm appreciative to each and every individual who helped me somehow for the satisfaction of this research work.

Abstract

Black holes is one of the fascinating object in Astrophysical world. In physical terms, a 'black hole' is a region where gravity is so strong that nothing- not even light- can escape. When a black hole is considered as a thermodynamic system, the laws of thermodynamics can be applied to the black hole. Now, clubbing laws of thermodynamics and black hole mechanics, the expression for different variables of the black hole can be derived and the thermodynamic stability of the black hole can be studied. In this project, I have considered solution of black hole in bumblebee gravity with a cosmological constant, which gives a component to the breaking of Lorentz symmetry by expecting a nonzero vacuum expectation value as an incentive for the bumblebee field. The case considered is when the effective cosmological constant is less than zero and greater than zero. For negative values of the cosmological constant, the black hole is in anti-de Sitter space and for positive values, in de-Sitter space. The different parameters of the black hole was found out by applying horizon conditions and the relation between entropy and area of the event horizon, the event horizon which is formed of space-time light rays is formed at the corners, the light rays neither go inside the black hole nor go outside, hence drifting in the corner constantly. The parameters of the black hole such as mass, temperature, free energy and heat capacity was derived and were plotted versus the entropy and the trend was observed. The thermodynamical stability was also studied by referring to the heat capacity.

Keywords – Entropy, Thermodynamics, Bumblebee gravity, Lorentz violating parameter, Heat Capacity

Contents

1 Introduction **2**

2 Background **3**
2.1 Thermodynamics . . . 3
2.1.1 Introduction to Thermodynamics . . . 3
2.1.2 Laws of Thermodynamics . . . 4
2.1.2.1 Zeroth Law of Thermodynamics . . . 4
2.1.2.2 First Law of Thermodynamics . . . 4
2.1.2.3 Second Law of Thermodynamics . . . 4
2.1.2.4 Third Law of Thermodynamics . . . 5
2.1.3 Definition of Entropy . . . 5
2.1.4 Thermodynamic Potentials . . . 5
2.1.5 Phase Transitions . . . 6
2.2 General Theory of Relativity . . . 7
2.3 Black Hole Thermodynamics . . . 10
2.3.1 Introduction to Black Hole Mechanics . . . 10
2.3.2 Laws of Black Hole Mechanics . . . 10
2.3.2.1 Zeroth Law of Black Hole Mechanics . . . 10
2.3.2.2 First Law of Black Hole Mechanics . . . 10
2.3.2.3 Second Law of Black Hole Mechanics . . . 10
2.3.2.4 Third Law Of Black Hole Mechanics . . . 11
2.3.2.5 Entropy and black hole . . . 11
2.3.2.6 Hawking Radiation . . . 12
2.3.2.7 Penrose process . . . 13

3 Literature Review **14**

4 Research Methodology **16**

5 Calculations **18**

6 Results and Discussion **20**

7 Conclusion and Future Scope **30**

References **31**

Appendix **32**
A1 Equations . . . 32
A2 Code . . . 34
A2.1 Anti-de Sitter . . . 34
A2.2 de-Sitter . . . 36

1 Introduction

Black Hole Thermodynamics is a fascinating concept whose foundation was laid by Jacob Bekenstein who proposed that Black holes have an entropy by considering black hole as a thermodynamical system. He also proposed No-Hair theorem which states that black holes are characterized by mass, spin or electric charge. Later came Stephen Hawking proposing that black holes evaporate and this phenomena is known as 'Hawking Radiation'. Bekenstein and Hawking put together a relation connecting entropy and area of event horizon known as Bekenstein-Hawking entropy. The laws of black hole mechanics is comparable to the laws of thermodynamics providing relation connecting area of event horizon and entropy.

In this work, primary objective is to find a relation connecting entropy to the mass, temperature, free energy and heat capacity of the black hole with a cosmological constant in bumblebee gravity and hence determine if the black hole is thermodynamically stable or not. Bumblebee gravity is said to possess gravitational fields and vector fields with non-zero vacuum expectation value, they are usually defined as the basic system for considering unconstrained breaking of Lorentz symmetry.

2 Background

2.1 Thermodynamics

2.1.1 Introduction to Thermodynamics

A thermodynamic system is a definite limit assigned to the matter for observation. Everything that remains outside this system is called surroundings or maybe another system as well.
System can be classified as-
1. Open system - if it can exchange mass, energy, or both between the system and surroundings.
2. Closed system - if it can only exchange energy but not mass with the surroundings.
3. Isolated system - if it does not allow the exchange of matter or energy with the surroundings.

State functions - It is the property of a physical quantity which depends on the state of the system. Ex: Mass, Pressure.
Path functions - It is the property of a physical quantity which depends on the path followed from initial to final. Ex: Heat, Work.

Extensive variables - A property that relies on the variables present on the system. Ex : Mass, Volume.
Intensive variables - A property that does not rely on the variables present on the system. Ex : Temperature, Density.

Thermodynamic equilibrium - The state of a system is in an equilibrium state if the macroscopic variables that describe the system don't change with respect to time.

2.1.2 Laws of Thermodynamics

2.1.2.1 Zeroth Law of Thermodynamics

Consider two systems A and B separated by an adiabatic wall, while both being in contact with another system C, along a diathermic wall. Both the systems A and B is said to be in thermal equilibrium with C after there occurs a change in the state of the systems. There will be no further change if adiabatic wall is replaced by diathermic wall or vice-versa once this is achieved. The thermodynamic variable that is equal in both system A and B is temperature since both are in thermal equilibrium.
Zeroth Law of Thermodynamics states that 'two systems in thermal equilibrium with a third system are in thermal equilibrium with each other' [1].

2.1.2.2 First Law of Thermodynamics

The first law of Thermodynamics can be more properly interpreted as the general law of conservation of energy which is applied to any system in which the transfer of energy from or to the surroundings is also considered.
First Law of Thermodynamics states that 'the energy supplied to the system not totally is used in doing work by the system on its surroundings rather it is also used to increase the internal energy of the system' [1].

2.1.2.3 Second Law of Thermodynamics

In simpler terms, the second law can be simplified as heat will spontaneously move from hot body to a cold body.It is a perception of the way that after some time contrasts in temperature, pressure and potential will in general level out in an actual system that is isolated from the surroundings. Entropy is a proportion of how much this interaction has advanced. The entropy of an isolated system that isn't in equilibrium will in general increment over the long run, moving toward a greatest estimation of equilibrium. The second law of thermodynamics sets up the idea of entropy as an actual property of a thermodynamic system.The second law of thermodynamics expresses that there exists in nature for every system of bodies a variable, which by all changes of the system either stays steady (in reversible processes) or expands in value (in irreversible processes). This variable is called, following Clausius, the **entropy** of the system [1].

2.1.2.4 Third Law of Thermodynamics

The third law defines what absolute zero is and bind it with the concepts of entropy and temperature. The third law can be elucidated as 'it is difficult to arrive at the absolute zero of temperature by any limited number of cycles' or 'the entropy of a perfect crystal approaches zero if temperature approaches absolute zero' [1].

2.1.3 Definition of Entropy

Entropy has various definitions - normally it can be defined as the state of randomness, uncertainty etc. Entropy is a property of a thermodynamic system that communicates the result of the instinctive changes in the system. Entropy is a proportion of a system's thermal energy for each unit temperature that is inaccessible for accomplishing useful work. In statistical mechanics, entropy is typically characterized as the proportion of various conceivable microscopic configurations Ω of the individual atoms and particles of the system which compares to the macroscopic state of the system.

$$S = k \ln \Omega \tag{2.1}$$

where k is the boltzmann constant and Ω is the number of microstates.
Entropy predicts that specific cycles are irreversible or inconceivable, regardless of not disobeying the conservation of energy.

2.1.4 Thermodynamic Potentials

Thermodynamic potentials are utilized to compute the energy of a system in terms of various factors but regularly we can just quantify certain properties of the system. It is usually used to define the thermodynamic state of the system. The five widely known thermodynamic potentials are -

1. Internal Energy(U) - it is the energy required to set up the system in any given internal state.

$$dU = Tds - PdV \tag{2.2}$$

   where T is the temperature, P is the pressure

2. Enthalpy(H) - it is a characteristic of a thermodynamic system described as the sum of the system's internal energy and the product of its volume and pressure.

$$H = U + PV \tag{2.3}$$

where U is the internal energy, P is the pressure and V is the volume.

3. Gibbs' Free energy(G) - it is a thermodynamic potential that can be utilized to determine the most extreme reversible work that might be performed by a thermodynamic system at constant temperature and pressure.

$$\Delta G = \Delta H - T\Delta S \tag{2.4}$$

where H is the enthalpy, T is the temperature and S is the entropy

4. Helmholtz Free Energy(F) - it is a thermodynamic potential that estimate the useful work available from a closed thermodynamic system at constant temperature and volume.

$$F = U - TS \tag{2.5}$$

where U is the internal energy, T is the temperature and S is the entropy

5. Landau Potential(Ω) - it is a quantity used in case of irreversible processes in open systems.

$$\Omega = U - TS - \mu N \tag{2.6}$$

U is the internal energy, T is the temperature, S is the entropy and μ is the chemical potential, and N is the number of particles in the system.

2.1.5 Phase Transitions

Phase transitions is described when a matter transforms from one phase to another. A phase of a thermodynamic system and the states of matter have uniform physical characteristics. In the course of a phase transition of a given medium, certain properties of the medium change, frequently spasmodically, because of the difference in outside conditions, like temperature, pressure, or others.

2.2 General Theory of Relativity

In his theory of special relativity, Albert Einstein confirmed that the laws of physics are identical among all non-accelerating up observers, and he demonstrated that the speed of light within a vacuum is the same regardless of how far an observer moves. As a result, he discovered that space and time were intertwined to form a single continuum known as space-time. Events that occur concurrently for one observer can occur at different times for another.

Postulates of Special Theory of relativity

1. The laws of physics are same in all inertial frame of reference.
2. The speed of light c is a constant, and unconcerned on the relative motion of the observer [2].

Special Theory wasn't able to explain the case with non-inertial frames of reference and the case with the bending of light near a massive body.

General Theory of relativity

The significant point set forward by the general theory of relativity is that the force of gravity emerged from a twisting of space time around a body of matter. In this manner a particle traveling through such a district of room follows a curved path, not a straight one.

After publishing special theory of relativity Einstein thought of bringing the concept of gravity to this and hence using Riemannian geometry, Einstein developed his model of defining gravity with his work.

The Einstein version of Equivalence principle, which was central to the general theory of gravity states that "In small enough areas of spacetime, the laws of physics lessen to those of special relativity, it is difficult to identify the presence of a gravitational field through local experiments". His thought was that if two lifts are considered, one rest at the Earth's surface, one speeding up in space. To an observer inside the lift (no windows) there is no experiment that he/she could perform to separate between the two situations [3].

Christoffel symbols

This is one of the most important equation which describes how the local coordinates bases gets transformed from one each point. The Christoffel connection is given by

$$\Gamma^{\sigma}_{\mu\nu} = \frac{1}{2} g^{\sigma\rho}(\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}) \tag{2.7}$$

The related association coefficients are now and again called Christoffel symbols and written as $\left\{ \begin{matrix} \sigma \\ \mu\nu \end{matrix} \right\}$. The investigation of manifolds with metric and their related associations is called Riemannian geometry, or now and again pseudo-Riemannian when the metric has a Lorentzian signature [3].

Free particles move along ways of "shortest conceivable distance" or *geodesics* - specifically which means particles make an honest effort to proceed onward straight lines, however in a curved spacetime there probably won't be any straight lines, so they do the following best thing. If $x^\mu(\lambda)$is the path followed by the particle, then the geodesic equation can be written as

$$\frac{d^2x^\mu}{d\lambda^2} + \Gamma^{\mu}_{\rho\sigma}\frac{dx^\rho}{d\lambda}\frac{dx^\sigma}{d\lambda} = 0 \tag{2.8}$$

Einstein field equations

The Einstein Field equations are written as,

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu} \tag{2.9}$$

where $G_{\mu\nu}$ is the Einstein tensor, $g_{\mu\nu}$ is the metric tensor, $T_{\mu\nu}$ is the stress–energy tensor, Λ is the cosmological constant and κ is the Einstein gravitational constant[2]. The Einstein tensor is given by

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} \tag{2.10}$$

Schwarzschild metric

In GR, the unique spherically symmetric vacuum solution is the Schwarzschild metric [3]. In spherical coordinates (t,r,θ,ϕ), the metric is given by,

$$ds^2 = -(1 - \frac{2GM}{r})dt^2 + (1 - \frac{2GM}{r})^{-1}dr^2 + r^2 d\Omega^2 \tag{2.11}$$

where G is the universal gravitational constant, M is the mass and r is the radius,
$d\Omega^2$ is the metric on a unit two-sphere,

$$d\Omega^2 = d\theta^2 + \sin^2\theta d\phi^2 \tag{2.12}$$

We discover from the above condition that the metric solution is singular when r= 2GM, this radius is called Schwarzschild radius r_S. It is static, spherically symmetric vacuum solution for Einstein's equation; M acts as a variable that can be explained as the conventional Newtonian mass that can be calculated by considering orbits at large distances from the gravitating source. *Schwarzschild radius* is the radius of the event horizon surrounding a stationary black hole.

$$r_S = \frac{2GM}{c^2} \tag{2.13}$$

where G is the universal gravitational constant.
c is the speed of light in vacuum.

2.3 Black Hole Thermodynamics

2.3.1 Introduction to Black Hole Mechanics

The four laws of black hole physics suggest about physical properties of black holes that are similar to thermodynamic laws. Brandon Carter, Stephen Hawking, and James Bardee were the people to find them. They proposed that the surface gravity of a black hole be identified with temperature and the area of the event horizon with entropy.

2.3.2 Laws of Black Hole Mechanics

2.3.2.1 Zeroth Law of Black Hole Mechanics

The zeroth law states that surface gravity is constant over an event horizon [4].(κ is constant.) We conclude that temperature and surface gravity are comparable.

2.3.2.2 First Law of Black Hole Mechanics

It states how the energy of a black hole changes with its horizon area, charge, and angular momentum [4].

$$dE = \frac{\kappa}{8\pi} dA + \Omega dJ + \phi dQ \tag{2.14}$$

where κ is surface gravity, A is the area of event horizon, Ω is the angular velocity, J is the angular momentum, ϕ is the electrostatic potential and Q is the electric charge.
Considering this with first law of thermodynamics, the energy conservation is being discussed.

2.3.2.3 Second Law of Black Hole Mechanics

The area of the event horizon of a black hole is a non-decreasing function.

$$\frac{dA}{dt} \geq 0 \tag{2.15}$$

This is Hawking's area theorem. As such, the area of a trapped surface can't diminish.Since the event horizon of a black hole is a particularly trapped surface, Hawking's area theorem follows inconsequentially[4].

2.3.2.4 Third Law Of Black Hole Mechanics

The third law of black hole mechanics states that it is impossible to let the surface gravity κ go to zero in a finite number of operations [4].
Therefore, surface gravity and entropy are constant at absolute zero.

2.3.2.5 Entropy and black hole

Entropy is identified with data (quantum bits). One normally ponders what the quantum pieces of black holes are for example what are the transporters of black hole data. Individuals have conjectured that the quantum pieces are some way or another disseminated over the event horizon, since $S \propto A$ [5].
The second law of thermodynamics gives change in entropy will consistently be a positive amount. In the event that we toss some matter having high entropy into the black hole, the total entropy of matter outside the black hole will diminish. In any case, one can say that the total entropy including the entropy inside the black hole won't diminish. Hence the area of the event horizon increases for a black hole. Following this Jacob Benkenstein recommended that the area of the event horizon is a proportion of the entropy of the black hole. As matter fall into a black hole, the total entropy outside the black hole and the area of the event horizon could never diminish. This forestalls the infringement of second law of thermodynamics [6].
Bekenstein Bound - The maximum amount of information that can be fit into any space - proportional to the surface area of that space.
Bekenstein and Hawking proposed a connection associating the entropy and the area of the event horizon of a black hole known as the Bekenstein Hawking equation.

$$S_{BH} = \frac{k_B A}{4 l_p^2} \tag{2.16}$$

$\rightarrow$k$_B$ is the boltzmann constant,
A is the area of the event horizon and
l_p is the Planck length = $\sqrt{\frac{G\hbar}{c^3}}$

2.3.2.6 Hawking Radiation

The second law of thermodynamics states that the entropy of an isolated system always increases, and that if two systems are joined together, then their total entropy will be greater than the sum of the entropies of the discrete systems. If black hole has an entropy, then it should also have a temperature and hence the body with temperature emits radiation. A black hole is said to emit particles and radiation since it is a hot body with a temperature which is in proportion with the mass - the higher the temperature, the lower the mass.

Stephen Hawking suggested that an empty space is not really empty. According to quantum theory, there are some particles coming from the event horizon that the black hole emits, also that empty means that all the fields including electromagnetic and gravitational should be zero. But, the uncertainty principle states that the more accurately one know the position of a particle, the less accurately one knows the other - since the rate of change of field with time and value of field is considered to be like velocity and position of a particle. There occurs some types of quantum fluctuations in the value of the field - this might be some particle pairs of light or gravity seen as together or apart or seen annihilate each other. These are virtual particles and cannot be detected, and since one of them is a real particle, the other have to be an antiparticle, one having positive energy and the other negative. The case with black hole is that the virtual particle with negative energy falls into the black hole and either becomes a real particle or an antiparticle. In this case, they don't have to annihilate with its partner, it's partner might well fall into the black hole or otherwise also escapes from the black hole as a real particle or an antiparticle [7].

By Einstein's mass-energy relation, $E = mc^2$, if a particle with negative energy falls into the black hole, its mass will be reduced, since energy and mass are proportional, the area of its event horizon also decreases as a result, hence entropy also decreases but this is compensated by the entropy of the emitted radiation, hence second law is not violated. The lower the mass, the higher the temperature. So, once black holes start losing it's mass, the temperature and rate of emission increases as a result the black hole mass decreases rapidly and once mass becomes so small, it is said to disappear completely in form of some emissions which can be compared to the explosion of millions of H-bombs.

Hawking Temperature is given by,

$$T_H = \frac{\hbar c^3}{8\pi G M k_B} \tag{2.17}$$

$\rightarrow \hbar$ is the reduced Planck's constant.

c is the speed of light in vacuum $(3x10^8 m/s)$.

G is the universal Gravitational constant.

M is the mass of the black hole and

k_B is the Boltzmann constant

2.3.2.7 Penrose process

For rotating black holes, apart from event horizon there appears to be an outer region of the event horizon known as ergosphere. So, Sir Roger Penrose showed that the energy of a rotating black can be extracted termed as Penrose process. The idea is that - if an object fall into Kerr black hole, it splits into two while in the ergosphere, one of them falls into the horizon with a negative energy but the other one will escapes into the external universe with more energy. This energy came from the black hole, nowhere else - Penrose process is said to decrease the angular momentum of a rotating black hole. Even though the mass of the black hole decreases, Penrose process doesn't violate area theorem - the area of event cannot decrease [3].

3 Literature Review

The bumblebee models are instances of proposition that include self-interfacing tensor fields with a nonzero vacuum expectation value(VEV) - which is the expectation value(average of all the achievable results) in the vacuum, the fields which characterize an advantaged heading in spacetime, in other words, they create an anisotropic stress-energy tensor. The least difficult case is depicted by a vector field B_ν and was first viewed as in the specific circumstance of string theories, with the Lorentz symmetry breaking set off by a smooth quadratic potential[8]. To make the model as reasonable as could be expected, the group set a black hole behind the scenes of a universe that was speeding up in its development (precisely like what we notice) and tuned the degree of evenness violation to coordinate with the conduct of dark energy that researchers measure.They tracked down that, for this situation, a black hole's shadow can show up to 10% more modest than it would in a normal-gravity world, giving an unmistakable method to test bumblebee gravity. While the current picture of black hole M87 is too fluffy to even think about differentiating, endeavors are in progress to take far better pictures of more black holes, testing the absolute most profound secrets of the universe simultaneously[8]. The action of bumblebee field is given by,

$$S_B = \int d^4x\sqrt{-g}[\frac{1}{2\kappa}(R-2\Lambda) + \frac{\xi}{2\kappa}B^\nu B^\mu R_{\mu\nu} - \frac{1}{4}B_{\mu\nu}B^{\mu\nu} - V(B^\nu B_\nu \pm b^2) + L_M] \quad (3.1)$$

where κ= $\frac{8\pi G}{c^4}$ is Einstein Gravitational constant, Λ is the cosmological constant, ξ is the coupling constant between Ricci tensor and bumblebee field, $B_{\mu\nu}$ is bumblebee field given as $B_{\mu\nu}$ = $\partial_\mu B_\nu$ - $\partial_\nu B_\mu$, V is the potential activating breaking of Lorentz symmetry and L_M represents the additional coupling with the bumblebee field[8].

de-Sitter and anti de-Sitter space

General Relativity predicts that space bends, since it is a physical thing, space can bend because of matter and energy, which makes the path of the objects moving through it curving. Curvature was introduced to explain the space-time geometry. For flat-regions of space, neither energy nor matter nearby, the objects moving along parallel paths continues to be in parallel paths. In positively curved regions of space, near planets or black holes, the parallel path converges, while that in negatively curved regions of space, the parallel path diverges. The maximally symmetric space-time with positive curvature ($\kappa > 0$) is called de-Sitter space - it is basically like a 3D

sphere with 2D projections. The maximally symmetric space-time with negative curvature ($\kappa < 0$) is called anti de-Sitter space - it is basically like a hyperboloid. The flat model which is with zero curvature ($\kappa = 0$) is known as Minkowski space, our universe is usually said to be flat[3]. In my case, assuming effective cosmological constant less than zero, the solution of the black hole is in anti de-Sitter space[8]. The event horizon of the spacetime is hence given by the radius,

$$r_+ = -\frac{1}{F(M,\Lambda_e,l)(1/3) - \frac{F(M,\Lambda_e,l)(1/3)}{(1+l)\Lambda_e}} \tag{3.2}$$

where $\Lambda_e = \frac{\kappa\lambda}{\xi}$ and $l = \xi b^2$ where b^2 is the radial component of the bumblebee field where,

$$F(M,\Lambda_e,l) = (1+l)^2\Lambda_e^2(3M + \sqrt{9M^2 - \frac{1}{(1+l)\Lambda_e}}) \tag{3.3}$$

When effective cosmological constant is assumed to greater than zero, then the black hole is in de-Sitter space, where the space-time has horizons of two - the event horizon and the cosmological horizon[8]. The effective cosmological constant should obey the inequality to come up with two horizons,

$$0 < \Lambda_e < \frac{1}{9M^2(1+l)} \tag{3.4}$$

The event horizons and cosmological horizon are expressed analytically as,

$$r_+ = \frac{2}{\sqrt{(1+l)\Lambda_e}} cos(\frac{\pi}{3} + \frac{\alpha}{3}) \tag{3.5}$$

$$r_c = \frac{2}{\sqrt{(1+l)\Lambda_e}} cos(\frac{\pi}{3} - \frac{\alpha}{3}) \tag{3.6}$$

with $\alpha = cos^{-1}(3M\sqrt{(1+l)\Lambda_e})$. The Lorentz-violating parameter l, changes the space-time structure, it diminishes the cosmological horizon range also, simultaneously, expands the event horizon range for significant greater value of Λ_e. The locale $r_+ < r < r_c$ is the supposed space of external correspondence. Around there, spectators may convey with one another, in this way there is no horizon around there

4 Research Methodology

If a black hole is regarded as a thermodynamic system, then the laws of thermodynamics can be applied to the black hole. Using laws of black hole mechanics and laws of thermodynamics, we can find parameters of any black hole and can observe if it is thermodynamically stable or not [9].

In this work, we take the metric equation of a black hole and there will be a f(r) function related with the metric equation. By putting f(r) to zero(applying Horizon condition), we can discover an equation for the mass of the black hole.

Entropy is given by,

$$S = \frac{A}{4} = \frac{4\pi r^2}{4} = \pi r^2 \tag{4.1}$$

So, radius in terms of entropy becomes

$$r = \sqrt{\frac{S}{\pi}} \tag{4.2}$$

Hence, mass can be written as consequence of entropy. In this way examining its variety.

By taking the derivative of the mass of the black hole, we get the temperature of the black hole given by,

$$T = \frac{\partial M}{\partial S} \tag{4.3}$$

Hence, plotting the temperature of the black hole with entropy, the trend can be observed.

The expression for free energy of the black hole is given by,

$$F = M - TS \tag{4.4}$$

where M is the mass of the black hole, T is the temperature and S is the entropy.

From thermodynamics, we find that heat capacity of the black hole is given by,

$$C = T\frac{\partial S}{\partial T} \tag{4.5}$$

Consequently we can discover mass, temperature, heat capacity and free energy of a black hole and break down the outcome graphically. From the heat capacity, we can analyse the thermodynamical stability of a black hole. For a thermodynamically stable system, it is found to be positive. For black holes, it is also observed to be negative. There also arises a case from which heat capacity shows a transition from negative to positive known as phase transition [9].

5 Calculations

The metric of the black hole with cosmological constant in bumblebee gravity is expressed as[8],

$$ds^2 = -f(r)dt^2 + (1+l)\frac{1}{f(r)}dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2) \tag{5.1}$$

where

$$f(r) = -(1 - \frac{2M}{r} - (1+l)\frac{\Lambda_e}{3}r^2) \tag{5.2}$$

l is the Lorentz violating parameter and Λ_e is the effective cosmological constant.

Mass of the black hole and the horizon radius can be related by applying the horizon condition as f(r) = 0.

Hence, the mass of the black hole is given by,

$$M = \frac{r}{2} - (1+l)\frac{\Lambda_e}{6}r^3 \tag{5.3}$$

But radius can be written in terms of entropy, hence, the mass of the black hole can be expressed as,

$$M = \frac{\sqrt{S}}{2\sqrt{\pi}} - (1+l)\frac{\Lambda_e}{6}\frac{S^{3/2}}{\pi^{3/2}} \tag{5.4}$$

Since we know that

$$r = \frac{\sqrt{S}}{\sqrt{\pi}}$$

Now, we can find other thermodynamic quantities in terms of entropy.

Temperature of the black hole is given by,

$$T = \frac{\partial M}{\partial S}$$

Free energy of the black hole is given by,

$$F = M - TS$$

Heat capacity of the black hole is given by,

$$C = T(\frac{\partial S}{\partial T})$$

The temperature of the black hole was found to be,

$$T = \frac{1}{4\sqrt{\pi S}} - \frac{(1+l)\Lambda_e\sqrt{S}}{4\pi^{3/2}} \tag{5.5}$$

The free energy was obtained as,

$$F = \frac{3\pi\sqrt{S} - \Lambda_e(1+l)S^{3/2}}{12\pi^{(3/2)}} \tag{5.6}$$

The heat capacity as a consequence of entropy, effective cosmological constant and Lorentz violating parameter was achieved as,

$$C_- = \frac{(\pi - (1+l)\Lambda_e S)}{4\pi^{3/2}\sqrt{S}} \frac{4\pi^2(\sqrt{4\pi^2T^2 - \Lambda_e(l+1)} + 2\pi T)^2}{\Lambda_e^2(l+1)^2\sqrt{4\pi^2T^2 - \Lambda_e(l+1)}} \tag{5.7}$$

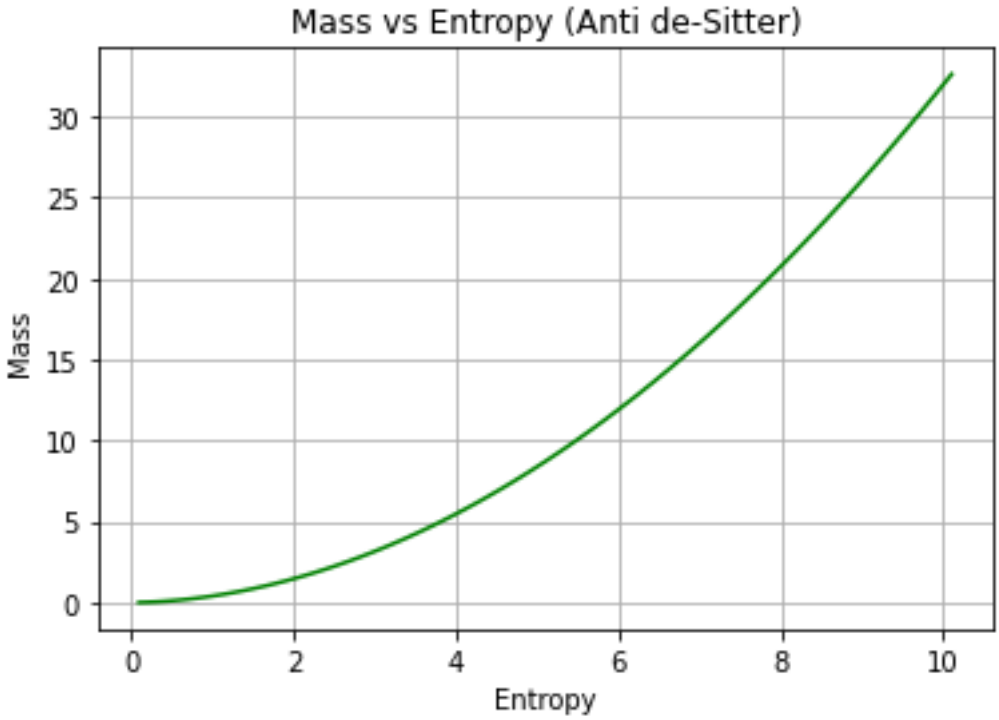
$$C_+ = \frac{(\pi - (1+l)\Lambda_e S)}{4\pi^{3/2}\sqrt{S}} \frac{-4\pi^2(\sqrt{4\pi^2T^2 - \Lambda_e(l+1)} - 2\pi T)^2}{\Lambda_e^2(l+1)^2\sqrt{4\pi^2T^2 - \Lambda_e(l+1)}} \tag{5.8}$$

6 Results and Discussion

The graphs were plotted for different parameters of the black hole v/s entropy and the results were as obtained.

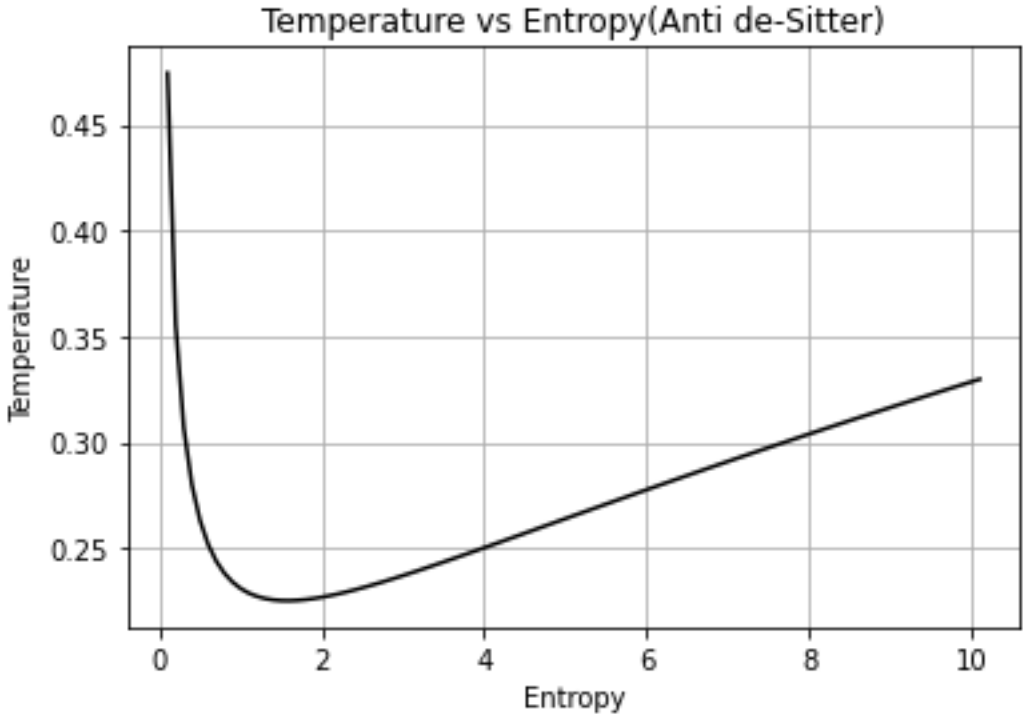
For black holes in Anti de-Sitter space

The plot for the mass of the black hole v/s entropy was found to be as shown below.



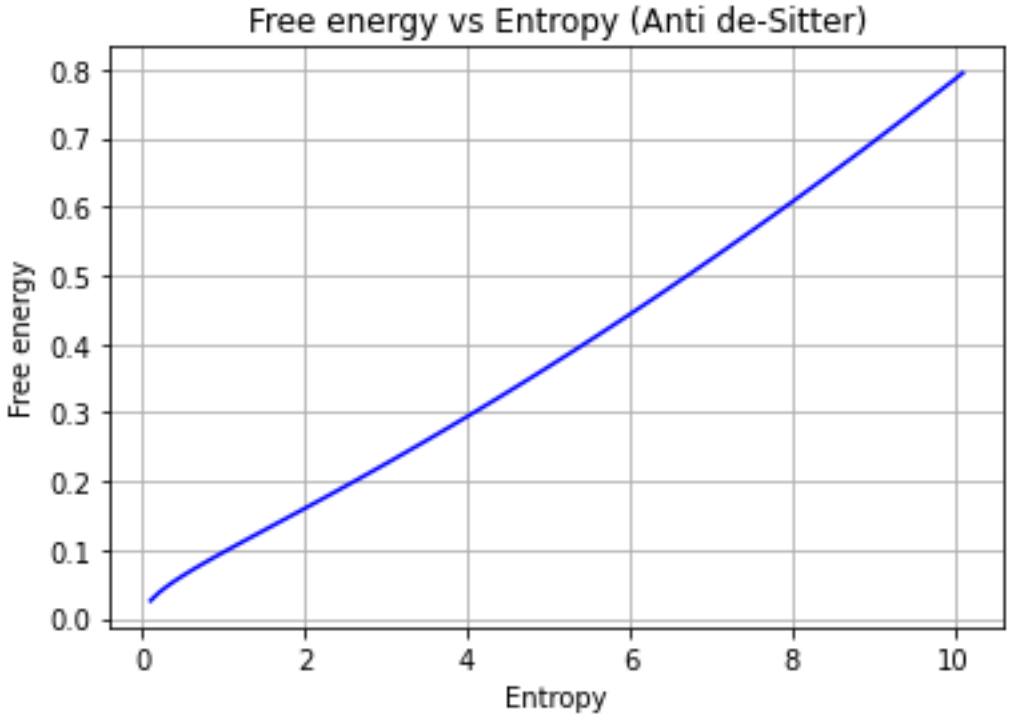
For Λ_e = -1 and l = 1, the trend is observed as - mass increases as entropy increases, which implies that area of event horizon also increases. Since we know that area law for entropy, as area increases, entropy also increases.

The plot for the temperature of the black hole v/s entropy was obtained as below.



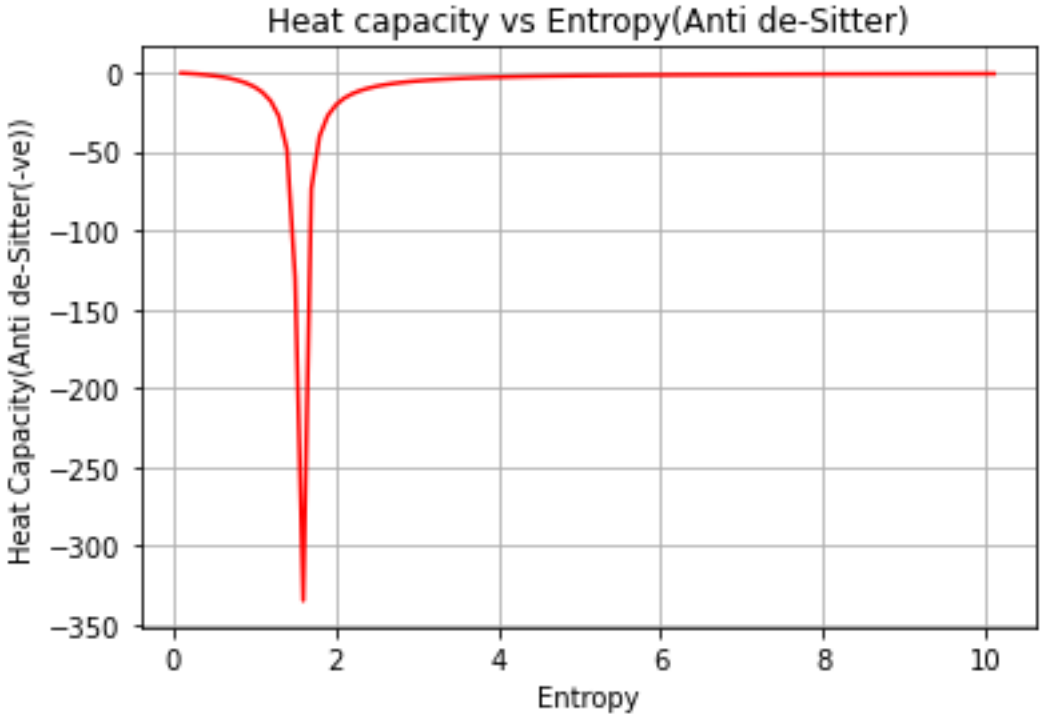
For Λ_e = -1 and l = 1, the trend is observed as - once entropy increases, there is a slight increase in the temperature of the black hole and goes on increasing as entropy rises. In other words, higher the temperature, higher the entropy.

The plot for the free energy of the black hole and entropy was acquired as shown.

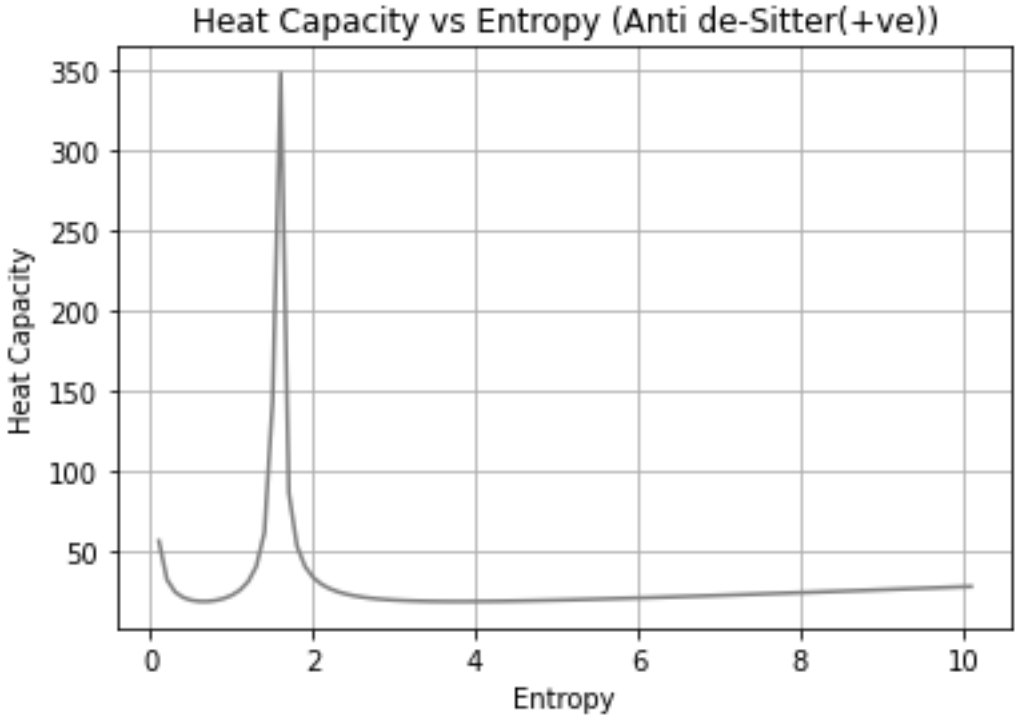


For Λ_e = -1 and l = 1, the trend is observed as - as entropy increases, the free energy of the black hole also increases. As thermodynamics suggests, free energy is a proportion of work a system can perform at constant temperature. hence higher the entropy means higher the work done by the system.

The graph between the heat capacity of the black hole and entropy is plotted below.



For Λ_e = -1 and l = 1, the trend is observed as - the heat capacity reaches -∞ and later on becomes zero as entropy rises, this is for one of the cases with entropy.



For the other case, the heat capacity falls to a minimum for an lower entropy and then goes on increasing. Hence, heat capacity increasing means higher the heat provided to cause a change in the temperature.

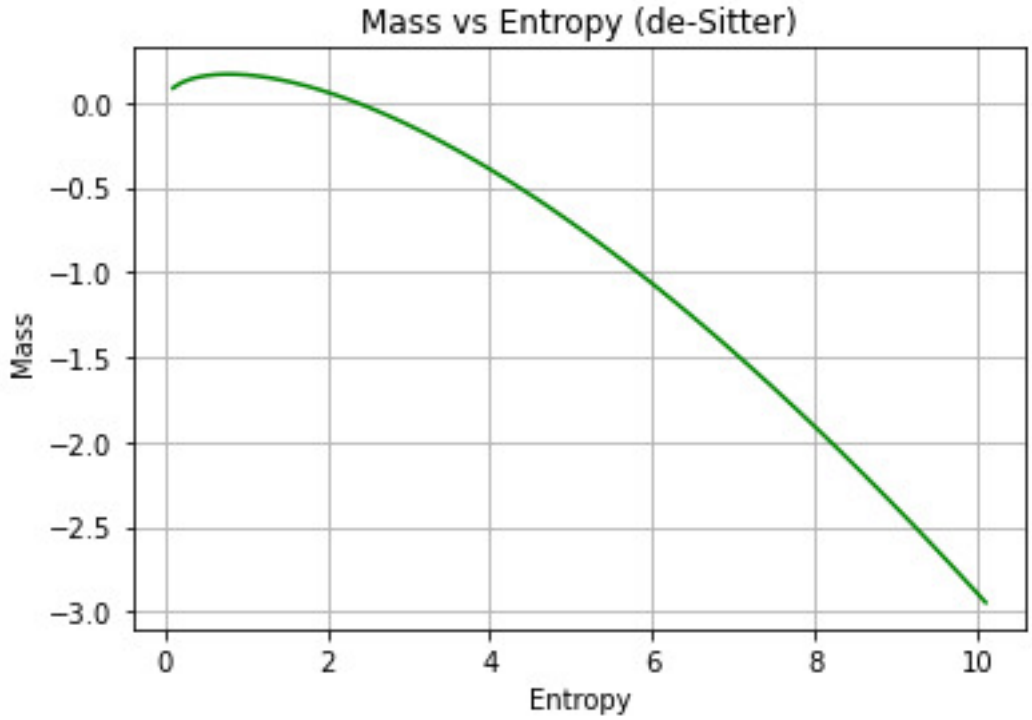
Heat capacity reaches infinity if either for $S = 0$ or

$$(l+1)^2 \Lambda_e^2 \sqrt{4\pi^2 \left(\frac{(\pi - (l+1)\Lambda_e S)}{4\pi^{\left(\frac{3}{2}\right)} \sqrt{S}} \right)^2 - \Lambda_e (l+1)} = 0 \tag{6.1}$$

This is also true for de-Sitter space .

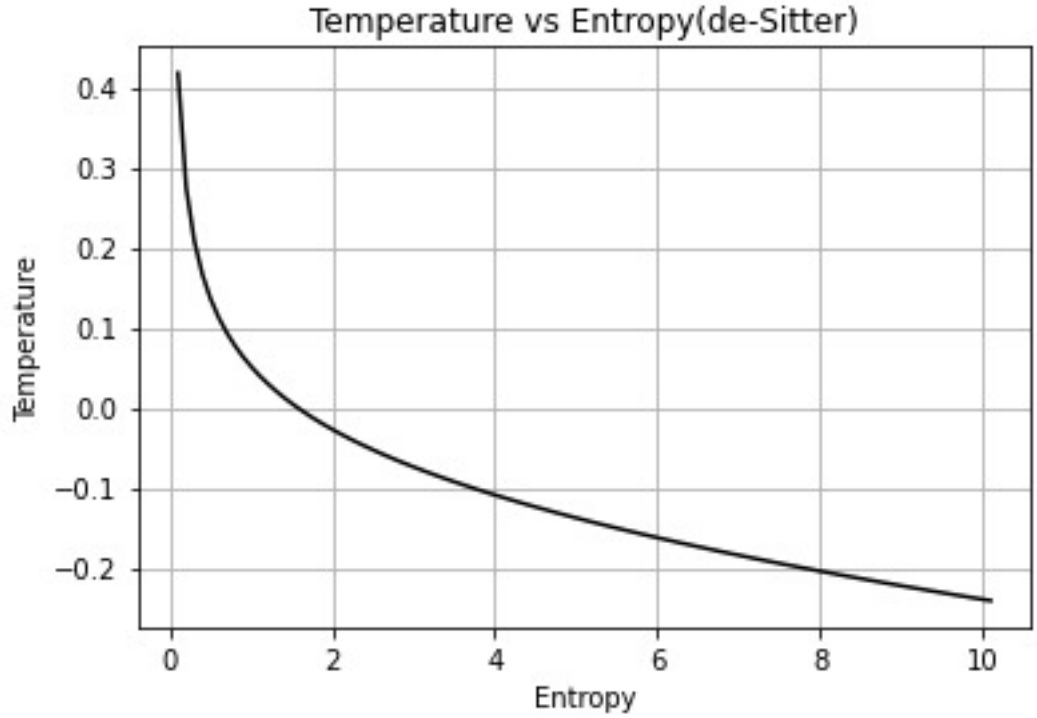
For black holes in de-Sitter space

The plot for the mass of the black hole v/s entropy was found to be as shown below.



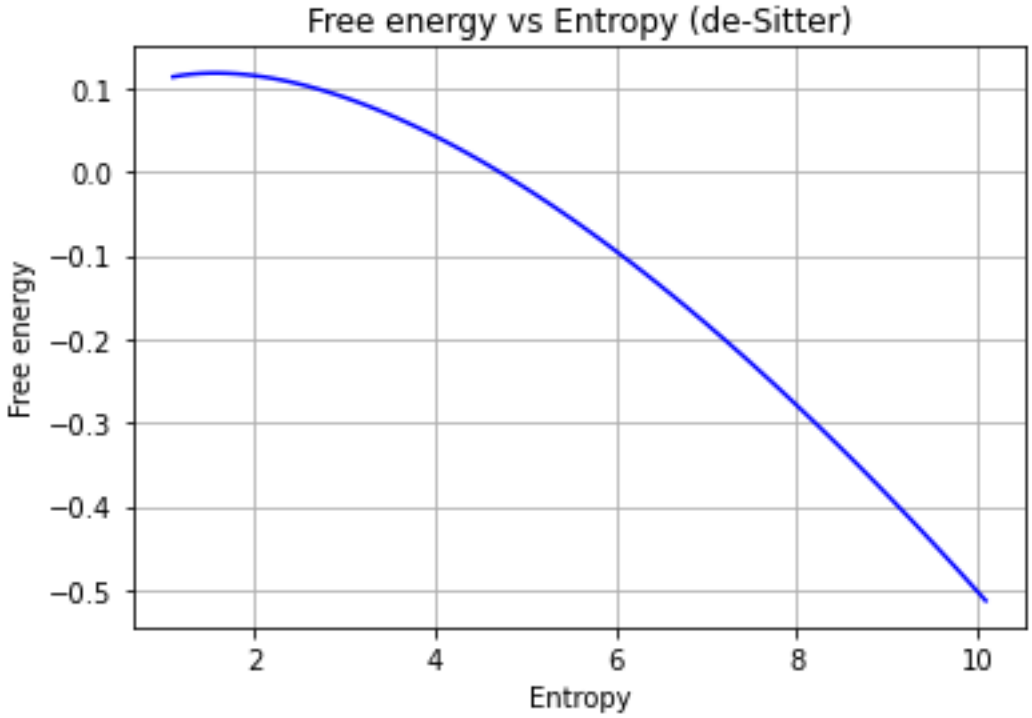
For $\Lambda_e = 1$ and $l = 1$, the trend is observed as - mass decreases as entropy increases. Mass can never be negative, this shows that the mass vs entropy in de-Sitter space is insignificant.

The plot for the temperature of the black hole v/s entropy was found to be as shown below.



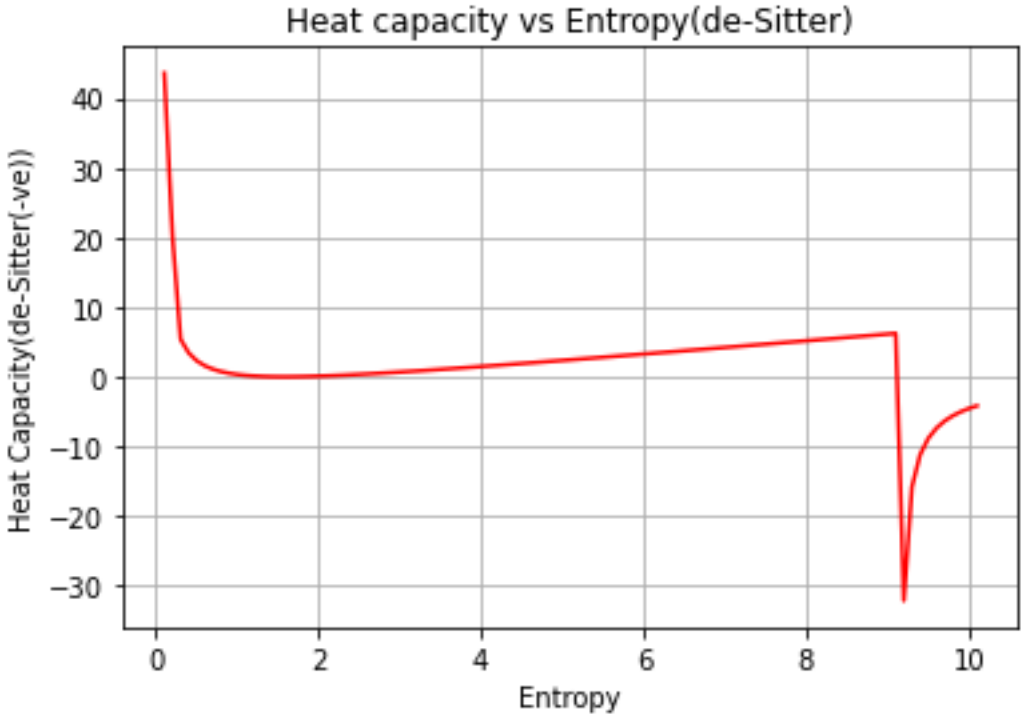
For $\Lambda_e = 1$ and $l = 1$, the trend is observed as - temperature decreases as entropy increases. This cannot happen, a temperature below absolute zero is not possible practically.

The plot for the free energy of the black hole v/s entropy was found to be as shown below.



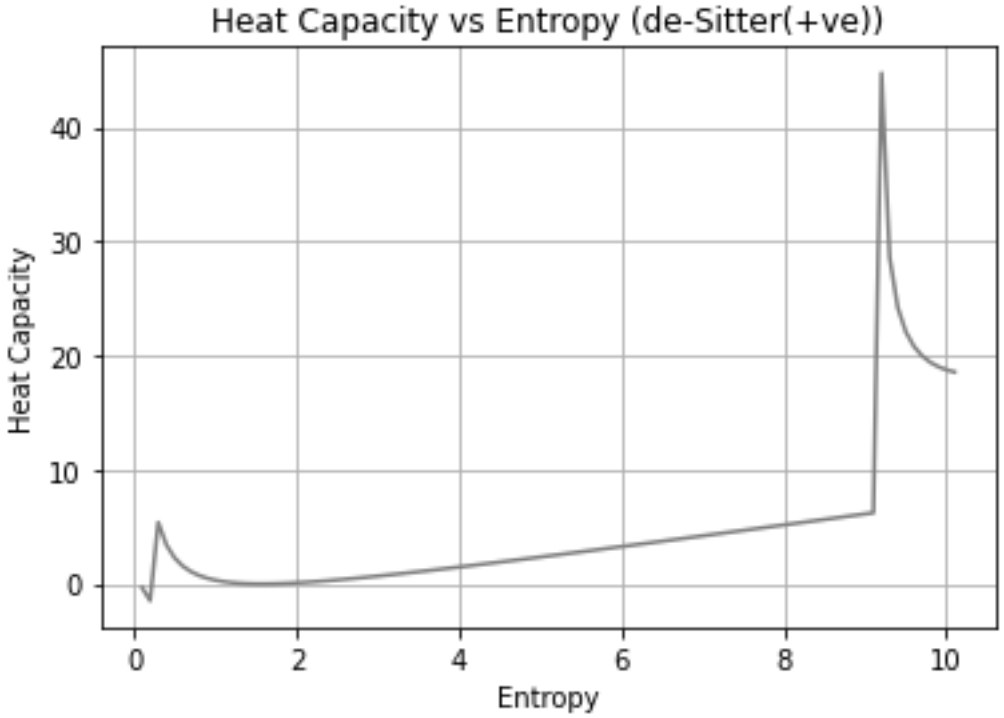
For $\Lambda_e = 1$ and $l = 1$, the trend is observed as - free energy decreases as entropy increases. This means that higher the entropy the lower the work done at constant temperature.

The graph was plotted for heat capacity vs entropy and the obtained graph is as shown



For $\Lambda_e = 1$ and $l = 1$, the trend is observed as - the heat capacity falls for a lower value of entropy and then on starts accelerating once entropy rises and after some value between 9 it again starts decreasing and becomes almost a constant value and then increases. This too shows a phase transition from positive to negative.

The second case for the heat capacity is shown below.



For Λ_e = 1 and l = 1, the trend is observed as - for a lower value of entropy, the heat capacity is observed to be of negative value then on as entropy increases, heat capacity also increases and at some point it becomes constant and then on decreasing. This graph shows discontinuity but a phase transition from negative to positive.

In the investigation of the thermodynamics of a black hole with cosmological constant in bumblebee gravity we examined the variation of temperature, mass and free energy of the black hole regarding its entropy and also noticed an intermittence in the heat capacity from which we summarizes that it goes through a phase transition.

7 Conclusion and Future Scope

In this work, I have determined the expressions for mass, free energy, temperature, and heat capacity of a black hole with cosmological constant in bumblebee gravity.The characterized changes for different parameters with entropy is plotted. What's more, we find that for mass v/s entropy graph, the thought of area law is justified. Heat capacity shows a sudden increase after the value of entropy rises. These are all summarized cases for a black hole in anti-de Sitter space.

In case of de-Sitter space, the black hole mass is showing some abnormal behaviour, mass is touching negative values for entropy, same is the case with temperature. Heat capacity shows an irregularity, which suggests that the black hole may go through a phase transition. So, we can conclude that a black hole in anti de-Sitter space, the black hole shows some stability, which we can summarize from our results but in case of de-Sitter space, we can't analyse anything regarding its parameters,since their graphs that obtained seems to be insignificant.

There were studies that the cosmological event horizon have temperature and an entropy which arised from the study of black hole thermodynamics and also utilizing laws of black hole mechanics there have studies on holographic principle which is fundamental to the AdS/CFT correspondance. There are additionally associations between black hole entropy and liquid surface tension.

References

[1] M. Planck, *Treatise on thermodynamics*. Courier Corporation, 2013.

[2] B. Schutz, *A first course in general relativity*. Cambridge university press, 2009.

[3] S. M. Carroll, *Spacetime and geometry*. Cambridge University Press, 2019.

[4] D. Lüst and W. Vleeshouwers, *Black hole information and thermodynamics*. Springer, 2019.

[5] S. W. Hawking, "Black holes and thermodynamics," *Physical Review D*, vol. 13, no. 2, p. 191, 1976.

[6] J. D. Bekenstein, "Black holes and entropy," in *JACOB BEKENSTEIN: The Conservative Revolutionary*. World Scientific, 2020, pp. 307–320.

[7] S. W. Hawking, "Particle creation by black holes," *Communications in mathematical physics*, vol. 43, no. 3, pp. 199–220, 1975.

[8] R. Maluf and J. C. Neves, "Black holes with a cosmological constant in bumblebee gravity," *Physical Review D*, vol. 103, no. 4, p. 044002, 2021.

[9] R. Tharanath and V. Kuriakose, "Thermodynamics and spectroscopy of schwarzschild black hole surrounded by quintessence," *Modern Physics Letters A*, vol. 28, no. 04, p. 1350003, 2013.

Appendix

A1 Equations

We find that

$$f(r) = (1 - \frac{2M}{r} - (1+l)\frac{\Lambda_e r^2}{3})$$

Equating, f(r) = 0, we get,

$$(1 - \frac{2M}{r} - (1+l)\frac{\Lambda_e r^2}{3}) = 0$$

So that mass of the black hole becomes,

$$M = \frac{r}{2} - (l+1)\frac{\Lambda_e r^3}{6}$$

in terms of entropy, mass can be written as,

$$M = \frac{\sqrt{S}}{2\sqrt{\pi}} - (1+l)\frac{\Lambda_e S^{3/2}}{6\pi^{3/2}}$$

Similarly, we know that temperature is given by,

$$T = \frac{\partial M}{\partial S}$$

so, we find that,

$$T = \frac{1}{4\sqrt{\pi S}} - \frac{(1+l)\Lambda_e\sqrt{S}}{4\pi^{3/2}}$$

and by rearranging, we get,

$$T = \frac{\pi - (1+l)\Lambda_e S}{4\pi^{3/2}\sqrt{S}}$$

For free energy of the black hole in terms of entropy, we know the expression,

$$F = M - TS$$

which means that,

$$F = \frac{\sqrt{S}}{2\sqrt{\pi}} - (1+l)\frac{\Lambda_e S^{3/2}}{6\pi^{3/2}} - (\frac{\pi - (1+l)\Lambda_e S}{4\pi^{3/2}\sqrt{S}})S$$

which can be simplified as ,

$$F = \frac{3\pi\sqrt{S} - S^{3/2}(1+l)\Lambda_e}{12\pi^{(3/2)}}$$

The case with the heat capacity is different, we know that

$$C = T\frac{\partial S}{\partial T}$$

but to derive with respect to S, we need to find the value of S from T. So, we know that,

$$T = \frac{\pi - (1+l)\Lambda_e S}{4\pi^{3/2}\sqrt{S}}$$

so, we find that entropy has two values since, we find that the expression for S is in the form of quadratic equation. Therefore,

$$\pi - (1+l)\Lambda_e S = T4\pi^{3/2}\sqrt{S}$$

Now, putting $S = x^2$, $\sqrt{S} = x$. Therefore, S is obtained as,

$$S = \frac{(4\pi^{3/2}T \pm (\sqrt{4\pi^{3/2}T)^2 - 4\Lambda_e(l+1)\pi})^2}{4\Lambda_e^2(l+1)^2}$$

Hence, we find C for two cases, for positive and negative,

For negative,

$$\frac{\partial S}{\partial T} = \frac{4\pi^2(\sqrt{4\pi^2T^2 - \Lambda_e(l+1)} + 2\pi T)^2}{\Lambda_e^2(l+1)^2\sqrt{4\pi^2T^2 - \Lambda_e(l+1)}}$$

For positive,

$$\frac{\partial S}{\partial T} = \frac{-4\pi^2(\sqrt{4\pi^2T^2 - \Lambda_e(l+1)} - 2\pi T)^2}{\Lambda_e^2(l+1)^2\sqrt{4\pi^2T^2 - \Lambda_e(l+1)}}$$

and hence, we get two cases for heat capacity,

$$C_+ = \frac{(\pi - (1+l)\Lambda_e S)}{4\pi^{3/2}\sqrt{S}} \frac{-4\pi^2(\sqrt{4\pi^2T^2 - \Lambda_e(l+1)} - 2\pi T)^2}{\Lambda_e^2(l+1)^2\sqrt{4\pi^2T^2 - \Lambda_e(l+1)}}$$

$$C_{-} = \frac{(\pi - (1+l)\Lambda_e S)}{4\pi^{3/2}\sqrt{S}} \frac{4\pi^2(\sqrt{4\pi^2T^2 - \Lambda_e(l+1)} + 2\pi T)^2}{\Lambda_e^2(l+1)^2\sqrt{4\pi^2T^2 - \Lambda_e(l+1)}}$$

Hence, the variations of entropy for both cases and also for both de-Sitter and anti de-Sitter cases.

A2 Code

The following codes were used in python to obtain the plots of the various variables of black hole with entropy.

A2.1 Anti-de Sitter

```
import numpy as np
import matplotlib.pyplot as plt
z=-1
j=1
p=np.pi
q=np.pi**2
r=np.pi**(3/2)
S=[]
T=[]
A=[]
C=[]
Cp=[]
F=[]
M=[]
i=1.5
while i<=20:
i=i+0.1
S.append(i)
x=(p-((j+1)*z*i))/(4*(r)*(i**(1/2)))
```

```
T.append(x)

a=(((4*q*(x**2))-z*(l+1))**(1/2))
A.append(a)

a1=(a+(2*p*x))**2
a1p=(a-(2*p*x))**2
a2=((j+1)**2)*(z**2)*a
y=-(x*4*q)*(a1)/(a2)
yp=-(x*4*q)*(a1p)/(a2)
C.append(y)
Cp.append(yp)

b1=(3*p*((i)**(1/2)))
b2=(z*(j+1)*((i)**(3/2)))
f=(b1-b2)/(12*r)
F.append(f)

c1=((i)**2)/(2*p**(1/2))
c2=((1+j)*z*(i**(3/2)))/(6*r)
m=c1-c2
M.append(m)

plt.figure(1)
plt.plot(S,T,color='black')
plt.grid()
plt.xlabel('Entropy')
plt.ylabel('Temperature')
plt.title('Temperature vs Entropy(Anti de-Sitter)')

plt.figure(2)
plt.plot(S,C,color='red')
plt.grid()
plt.xlabel('Entropy')
plt.ylabel('Heat Capacity(Anti de-Sitter(-ve))')
plt.title('Heat capacity vs Entropy(Anti de-Sitter)')
```

```
plt.figure(3)
plt.plot(S,Cp,color='grey')
plt.grid()
plt.xlabel('Entropy')
plt.ylabel('Heat Capacity')
plt.title('Heat Capacity vs Entropy (Anti de-Sitter(+ve))')

plt.figure(4)
plt.plot(S,F,color='blue')
plt.grid()
plt.xlabel('Entropy')
plt.ylabel('Free energy')
plt.title('Free energy vs Entropy (Anti de-Sitter)')

plt.figure(5)
plt.plot(S,M,color='green')
plt.grid()
plt.xlabel('Entropy')
plt.ylabel('Mass')
plt.title('Mass vs Entropy (Anti de-Sitter)')
```

A2.2 de-Sitter

```
import numpy as np
import matplotlib.pyplot as plt
z=1
j=1
p=np.pi
q=np.pi**2
r=np.pi**(3/2)
S=[]
T=[]
A=[]
C=[]
```

```
Cp=[]
F=[]
M=[]
i=0
while i<=10:
i=i+0.1
S.append(i)
x=(p-((j+1)*z*i))/(4*(r)*(i**(1/2)))
T.append(x)

a=(((4*q*(x**2))-z*(l+1))**(1/2))
A.append(a)

a1=(a+(2*p*x))**2
a1p=(a-(2*p*x))**2
a2=((j+1)**2)*(z**2)*a
y=-(x*4*q)*(a1)/(a2)
yp=-(x*4*q)*(a1p)/(a2)
C.append(y)
Cp.append(yp)

b1=(3*p*((i)**(1/2)))
b2=(z*(j+1)*((i)**(3/2)))
f=(b1-b2)/(12*r)
F.append(f)

c1=((i)**(1/2))/(2*p**(1/2))
c2=((1+j)*z*(i**(3/2)))/(6*r)
m=c1 - c2
M.append(m)

plt.figure(1)
plt.plot(S,T,color='black')
plt.grid()
plt.xlabel('Entropy')
```

```
plt.ylabel('Temperature')
plt.title('Temperature vs Entropy(de-Sitter)')

plt.figure(2)
plt.plot(S,C,color='red')
plt.grid()
plt.xlabel('Entropy')
plt.ylabel('Heat Capacity(de-Sitter(-ve))')
plt.title('Heat capacity vs Entropy(de-Sitter)')

plt.figure(3)
plt.plot(S,Cp,color='grey')
plt.grid()
plt.xlabel('Entropy')
plt.ylabel('Heat Capacity')
plt.title('Heat Capacity vs Entropy (de-Sitter(+ve))')

plt.figure(4)
plt.plot(S,F,color='blue')
plt.grid()
plt.xlabel('Entropy')
plt.ylabel('Free energy')
plt.title('Free energy vs Entropy (de-Sitter)')

plt.figure(5)
plt.plot(S,M,color='green')
plt.grid()
plt.xlabel('Entropy')
plt.ylabel('Mass')
plt.title('Mass vs Entropy (de-Sitter)')
```